39 The charge-to-mass ratio *r* of a particle is given by the equation shown.

$$r = \frac{\text{charge on particle}}{\text{mass of particle}}$$

The value of *r* is determined for an α-particle, a β^+ particle and a proton p.

Which list shows the particles in order of increasing magnitude of *r* from left to right?

A $\alpha \rightarrow \beta^+ \rightarrow \text{p}$

B $\alpha \rightarrow \text{p} \rightarrow \beta^+$

C $\text{p} \rightarrow \alpha \rightarrow \beta^+$

D $\text{p} \rightarrow \beta^+ \rightarrow \alpha$